Linear Algebra

Linear/Vector Space

Vector Space: A set V over field $\mathbb{R}$ is a vector space if:

- $\forall \mathbf{a}, \mathbf{b} \in V, \mathbf{a} + \mathbf{b} \in V$ (closure under addition)
- $\forall \mathbf{a} \in V, c \in \mathbb{R}, c \cdot \mathbf{a} \in V$ (closure under scalar multiplication)

Basis Vectors: A set $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ is a basis if:

- $\forall \mathbf{v} \in V, \exists \lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{R}$ such that $\mathbf{v} = \sum_{i=1}^{n} \lambda_i \mathbf{e}_i$
- The vectors are linearly independent

Inner/Dot/Scalar Product:

- $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^T \mathbf{b} = \sum_{i=1}^{n} a_i b_i$
- Properties: symmetric, linear, positive definite

Outer Product:

- $\mathbf{C} = \mathbf{a}\mathbf{b}^T$ where $C_{ij} = a_i b_j$
- Results in a matrix of rank 1

Orthogonality:

- Vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal if $\langle \mathbf{a}, \mathbf{b} \rangle = 0$
- Orthonormal basis: $\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij}$

Vector Norms:

- l_p-norm: $\|\mathbf{a}\|_p = \left(\sum_{i=1}^{n} |a_i|^p\right)^{1/p}$
- l_1-norm (Manhattan): $\|\mathbf{a}\|_1 = \sum_{i=1}^{n} |a_i|$
- l_2-norm (Euclidean): $\|\mathbf{a}\|_2 = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle}$
- l_∞-norm (Maximum): $\|\mathbf{a}\|_\infty = \max(|a_1|, |a_2|, \ldots, |a_n|)$

Matrix Operations

Linear Transformation: A mapping $T : V \to W$ is linear if:

- $T(\mathbf{a} + \mathbf{b}) = T(\mathbf{a}) + T(\mathbf{b})$
- $T(c \cdot \mathbf{a}) = c \cdot T(\mathbf{a})$
- Every linear transformation can be represented by matrix multiplication

Affine Transformation:

- $\mathbf{y} = M\mathbf{x} + \mathbf{b}$
- Combination of linear transformation and translation

Matrix Addition:

- $\mathbf{C} = \mathbf{A} + \mathbf{B}$ where $C_{ij} = A_{ij} + B_{ij}$
- Requires matrices of same dimensions

Matrix Multiplication:

- $\mathbf{C} = \mathbf{AB}$ where $C_{ij} = \sum_k A_{ik}B_{kj}$
- Number of columns of $\mathbf{A}$ must equal number of rows of $\mathbf{B}$

Hadamard Product (Element-wise):

- $\mathbf{C} = \mathbf{A} \circ \mathbf{B}$ where $C_{ij} = A_{ij}B_{ij}$
- Requires matrices of same dimensions

Kronecker Product:

- $\mathbf{C} = \mathbf{A} \otimes \mathbf{B}$ where $C_{ik,jl} = A_{ij}B_{kl}$
- Block matrix construction

Trace:

- $tr(\mathbf{A}) = \sum_i A_{ii}$
- Properties: $tr(\mathbf{A} + \mathbf{B}) = tr(\mathbf{A}) + tr(\mathbf{B})$, $tr(\mathbf{AB}) = tr(\mathbf{BA})$

Determinant:

- $det(\mathbf{A}) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} A_{i,\sigma(i)}$
- Geometric interpretation: scaling factor of linear transformation
- $det(\mathbf{AB}) = det(\mathbf{A})det(\mathbf{B})$

Matrix Rank:

- Dimension of column/row space
- $rank(\mathbf{AB}) \leq \min(rank(\mathbf{A}), rank(\mathbf{B}))$
- Full rank: $rank(\mathbf{A}) = \min(m, n)$

Matrix Norms:

- p-norm: $\|\mathbf{A}\|_p = \sup_{\mathbf{x} \neq 0} \frac{\|\mathbf{Ax}\|_p}{\|\mathbf{x}\|_p}$

- Frobenius norm: $\|\mathbf{A}\|_F = \left(\sum_m \sum_n |A_{mn}|^2\right)^{1/2}$

Special Matrices

Diagonal Matrix:

- $A_{ij} = 0$ if $i \neq j$
- Denoted as $diag(\lambda_1, \lambda_2, \ldots, \lambda_n)$

Identity Matrix:

- $I_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$
- $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$

Symmetric Matrix:

- $\mathbf{A} = \mathbf{A}^T$ $(A_{ij} = A_{ji})$
- Real symmetric matrices have real eigenvalues and orthogonal eigenvectors

Orthogonal Matrix:

- $\mathbf{AA}^T = \mathbf{A}^T\mathbf{A} = \mathbf{I}$
- Columns form orthonormal basis
- Preserves lengths and angles: $\|\mathbf{Ax}\|_2 = \|\mathbf{x}\|_2$

Positive Definite Matrix:

- $\mathbf{x}^T\mathbf{Ax} > 0$ for all $\mathbf{x} \neq \mathbf{0}$
- All eigenvalues are positive
- Symmetric positive definite matrices have Cholesky decomposition

Sparse Matrix:

- Majority of elements are zero
- Efficient storage and computation

Matrix Decompositions

Eigen Decomposition:

- $\mathbf{A} = \mathbf{Q\Lambda Q}^{-1}$ where $\mathbf{Aq}_i = \lambda_i \mathbf{q}_i$
- $\mathbf{Q}$ contains eigenvectors, $\mathbf{\Lambda}$ contains eigenvalues
- Applicable to diagonalizable matrices

Singular Value Decomposition (SVD):

- $\mathbf{A} = \mathbf{U\Sigma V}^T$ where $\mathbf{Av}_i = \sigma_i \mathbf{u}_i$

- $\mathbf{U}$, $\mathbf{V}$ orthogonal, $\mathbf{\Sigma}$ diagonal with singular values
- Always exists for any matrix
- Applications: PCA, low-rank approximation, pseudoinverse

Other Important Decompositions:

- **QR Decomposition**: $\mathbf{A} = \mathbf{QR}$ (orthogonal × upper triangular)
- **LU Decomposition**: $\mathbf{A} = \mathbf{LU}$ (lower × upper triangular)
- **Cholesky Decomposition**: $\mathbf{A} = \mathbf{LL}^T$ for positive definite matrices